

EC9540 - HUMAN COMPUTER INTERACTION

Lab 02

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MediLink: A Community Healthcare Information and Support Platform

Introduction

Healthcare services are essential for improving people's quality of life. However, many patients still face difficulties when accessing hospital services due to manual appointment booking, limited information about doctor and medicine availability and time-consuming payment processes. These challenges often lead to unnecessary hospital visits, long waiting times and inconvenience for patients.

To address these issues, we propose **MediLink**, a web-based healthcare platform designed to improve the patient experience through a simple and user-friendly interface. Based on the findings of our user requirements survey, the scope of the system focuses on three core patient services:

- Checking doctor availability and booking appointments.
- Checking medicine availability before visiting the hospital.
- Making secure online payments while providing patient details.

These features were selected because they directly address the main problems identified through the questionnaire survey. They also provide a complete and convenient user journey, allowing patients to check doctor availability, book appointments, verify medicine availability and complete payments online without unnecessary hospital visits. The system is designed following Human-Computer Interaction (HCI) principles to ensure usability, efficiency and a positive user experience.

Target Users

The **MediLink** platform is designed to support patients in accessing hospital services more efficiently while also assisting doctors and hospital staff in managing healthcare services. The system is designed with a user-centered approach, ensuring that each user group can complete their tasks through a simple, efficient and easy-to-use interface.

The main target users of the system are patients, doctors and hospital staff. In addition, parents, guardians and elderly users are considered secondary users because they use the same patient interface under different situations.

Primary User: Patients

Patients are the main users of the MediLink platform. They include students, working adults, senior citizens and any individual seeking healthcare services. The system helps patients complete essential hospital-related tasks without unnecessary hospital visits.

Using MediLink, patients can:

- Search for doctors by hospital or medical specialty.
- Check doctor availability.
- Book, reschedule or cancel appointments.
- Check medicine availability before visiting the hospital.

- Enter personal and appointment details.
- Make secure online payments.
- View appointment and payment confirmations.

The platform helps patients save time, reduce waiting periods and improve access to healthcare services through a simple and user-friendly interface.

Secondary User: Doctors

Doctors use MediLink to manage their consultation schedules and appointments efficiently. Instead of handling appointments manually, doctors can keep their availability up to date and review scheduled appointments.

Doctors can:

- Manage their availability.
- View daily appointment schedules.
- Accept or update appointment information.
- Monitor booked patients.

The system helps doctors organize their schedules more efficiently and reduce administrative workload.

Secondary User: Hospital Staff

Hospital staff are responsible for maintaining hospital information and ensuring that patients receive accurate and up-to-date information through the system.

Hospital staff can:

- Manage doctor information.
- Update doctor availability.
- Update medicine availability.
- Verify appointment details.
- Confirm online payments.
- Manage hospital service information.

The system reduces manual administrative work while improving the accuracy and reliability of hospital information.

Additional User Groups

Parents and Guardians

Parents or guardians can use MediLink on behalf of children or dependent family members. They can search for doctors, book appointments, check medicine availability and complete online payments using the same patient interface.

Elderly Users

Older adults may use the platform independently or with assistance from family members. Therefore, the interface should be simple, easy to navigate and accessible. Features such as large buttons, clear labels, readable text and a minimal number of steps improve usability for elderly users.

Summary of Questionnaire Findings

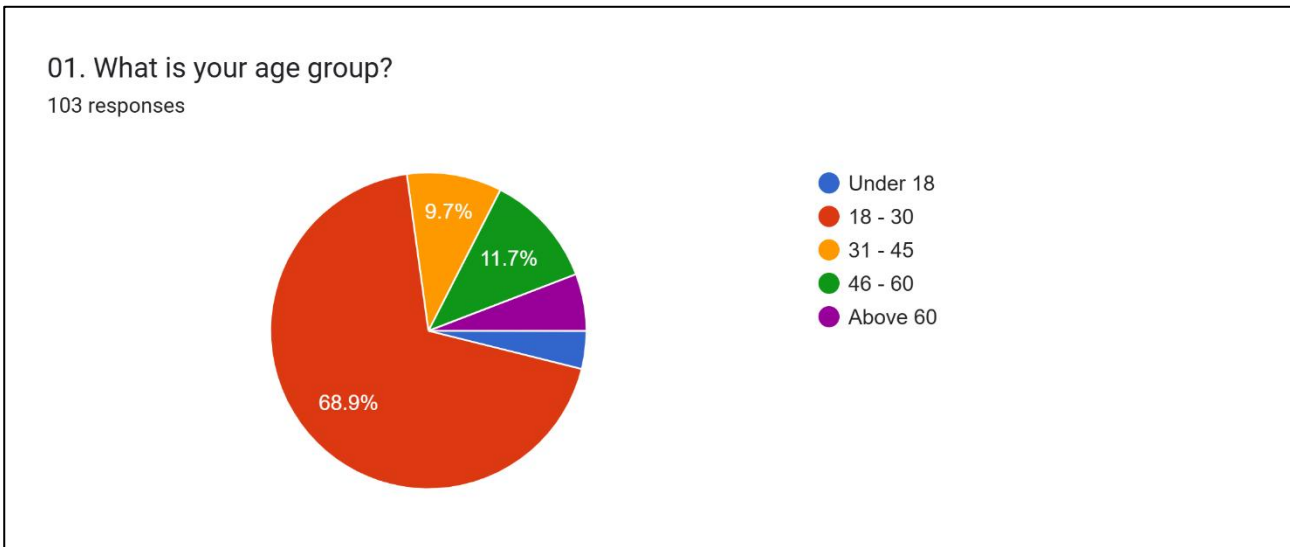
This section summarizes the final questionnaire results relevant to the revised, narrowed scope of MediLink, based on the complete set of 103 responses collected. The complete response set is available in the survey analytics report; the figures below are drawn directly from that data.

Respondent Profile and Digital Readiness

The sample is heavily weighted toward younger adults: the 18 - 30 age group alone accounts for more than two-thirds of all respondents, as summarized in Table 1.

Age Group	Share of Respondents
18 - 30	68.9%
46 - 60	11.7%
31 - 45	9.7%
Under 18 and Above 60 (combined)	≈ 9.7%

Table 1: Age-group distribution of respondents (n = 103)



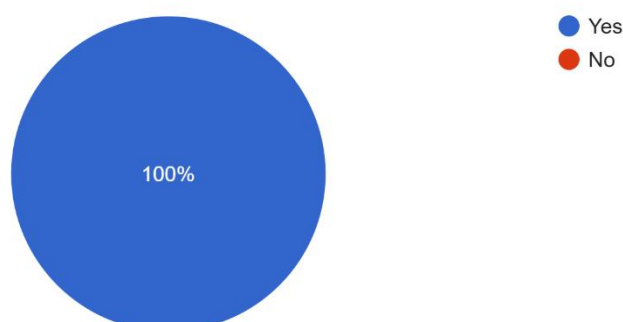
This age profile is an important design consideration: while the majority of respondents are digitally confident younger adults, a meaningful minority fall into the elderly bracket identified as a secondary user group, reinforcing the requirement for a simple, accessible and readable interface rather than one designed only for a digitally fluent audience.

Aspect	Key Finding
Smartphone ownership	All respondents (100%) reported owning a smartphone.
Internet usage frequency	The large majority (87.4%) use the Internet daily, with the remaining respondents split across weekly, rarely and never use still a small minority overall, but a modest increase in occasional users compared with the earlier data pull.
Comfort using healthcare websites	94.2% of respondents were “very comfortable” (64.1%) or “comfortable” (30.1%) using websites for healthcare services; the remainder were neutral and none reported discomfort.
Hospital visits frequency	Most respondents visit a hospital monthly, with a further share visiting every few months; weekly visits are comparatively uncommon.

Table 2: Respondent digital profile (n = 103)

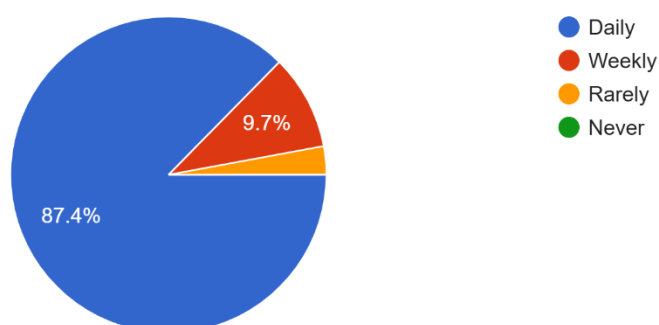
02. Do you own a smartphone?

103 responses



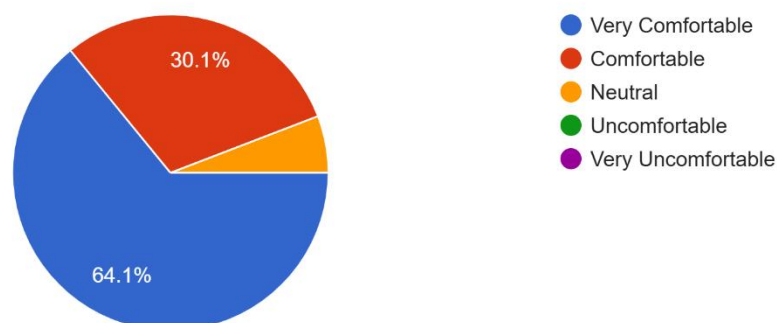
03. How often do you use the Internet?

103 responses



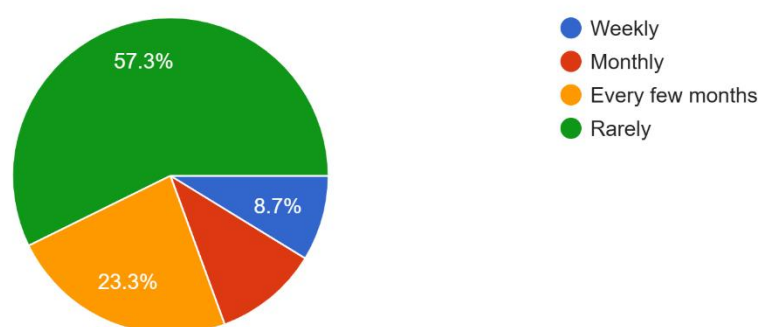
15. How comfortable are you using websites to access healthcare services?

103 responses



04. How often do you visit a hospital?

103 responses



Current Appointment Booking Experience

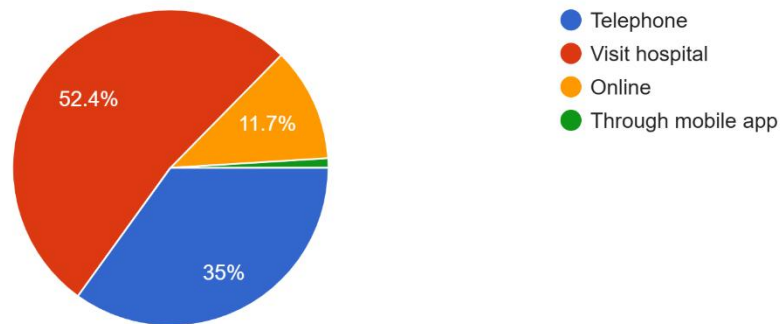
Most respondents still rely on traditional channels visiting the hospital in person or booking by telephone to secure an appointment, while online booking accounts for a smaller share of current behavior (11.7%). This confirms both a persistent gap in digital adoption and a clear opportunity for MediLink to shift booking behavior online.

Satisfaction with the current appointment booking process remains low: 40.8% of respondents described themselves as very dissatisfied, with a further share dissatisfied or neutral, and only a minority satisfied overall. In parallel, many respondents reported experiencing long waiting times at hospitals either “always” or “often”, and 44.7% said they had personally experienced difficulty obtaining a doctor's appointment, most commonly citing hospital overcrowding, long waiting times and busy phone lines as the underlying causes.

When asked how useful an online appointment booking system would be, 95.1% of respondents rated it “very useful” or “useful”, an even stronger and more decisive signal than in the earlier data pull that this feature should be a core part of the platform.

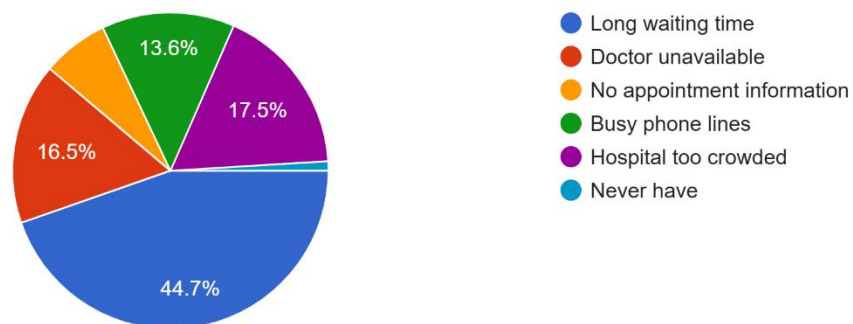
05. How do you currently book hospital appointments?

103 responses



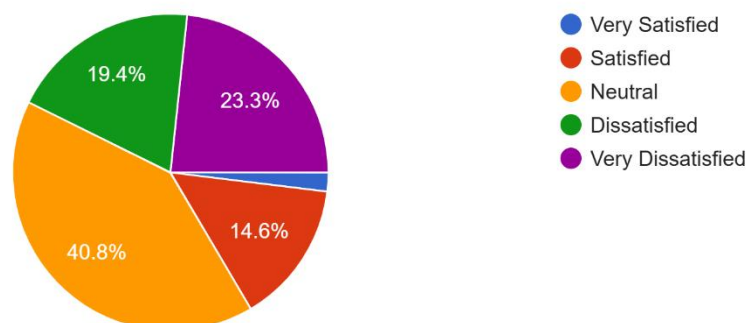
06. Have you ever experienced difficulty getting a doctor's appointment?

103 responses



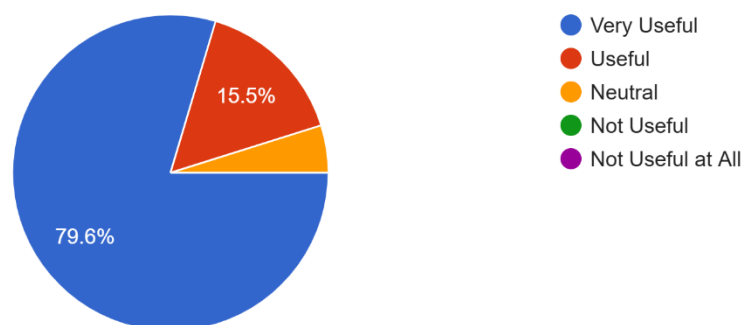
07. How satisfied are you with the current appointment booking process?

103 responses



10. How useful would an online appointment booking system be?

103 responses

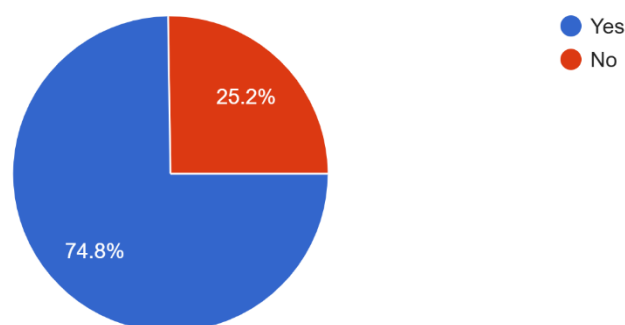


Medicine Availability

25.2% of respondents reported having travelled to a hospital only to discover that the medicine they required was unavailable an avoidable inconvenience directly addressed by the second core feature of the revised scope. Correspondingly, 92.2% of respondents rated the ability to check medicine availability before visiting the hospital as “very useful” or “useful”, confirming it as one of the most strongly endorsed features in the questionnaire, second only to appointment booking.

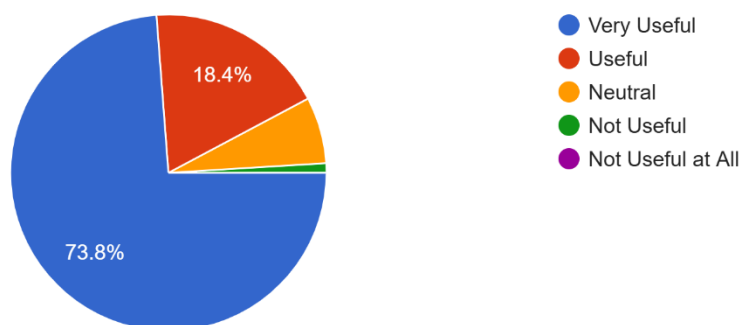
09. Have you ever visited a hospital only to find that your required medicine was unavailable?

103 responses



11. How useful would checking medicine availability before visiting the hospital be?

103 responses

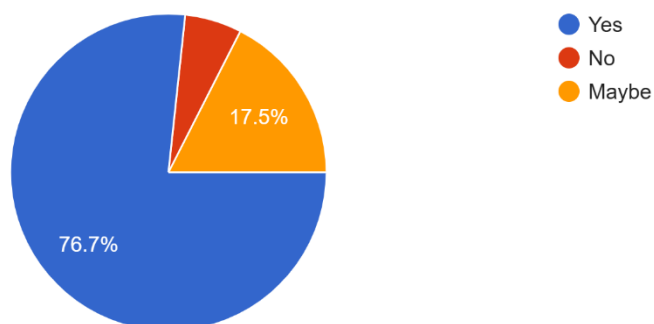


Willingness to Pay Online

76.7% of respondents indicated they would be willing to pay for hospital appointments online, with a further 17.5% saying “maybe” and only a small minority declining outright. Combined with the finding that secure payment was rated an important website attribute by 64.1% of respondents, this provides strong support for including online payment as a core, rather than optional, feature of the initial release.

12. Would you be willing to pay for hospital appointments online?

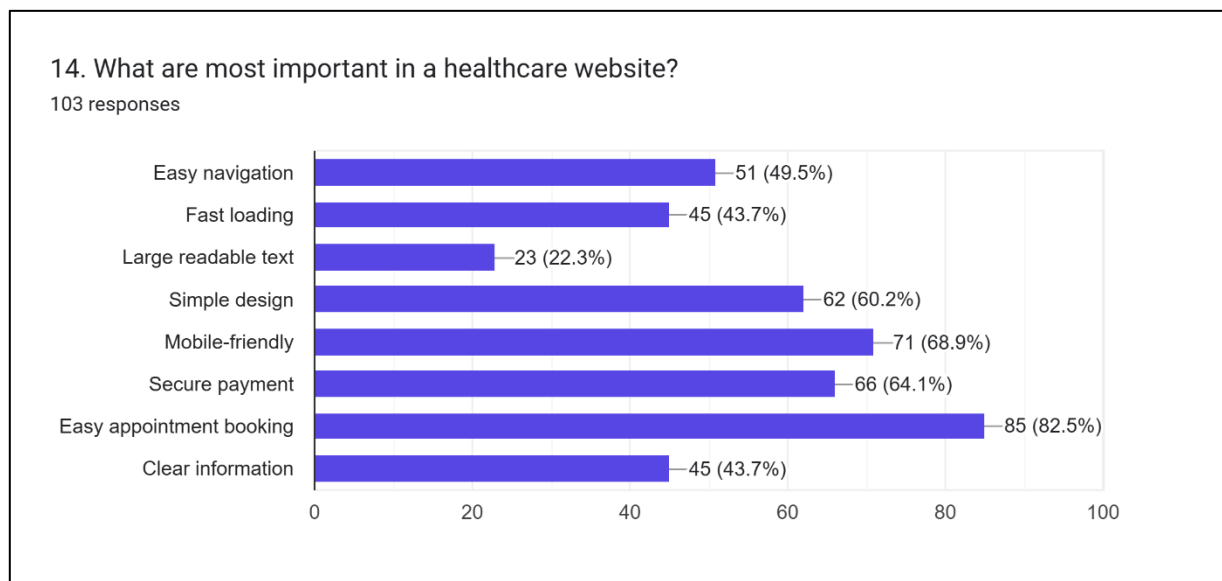
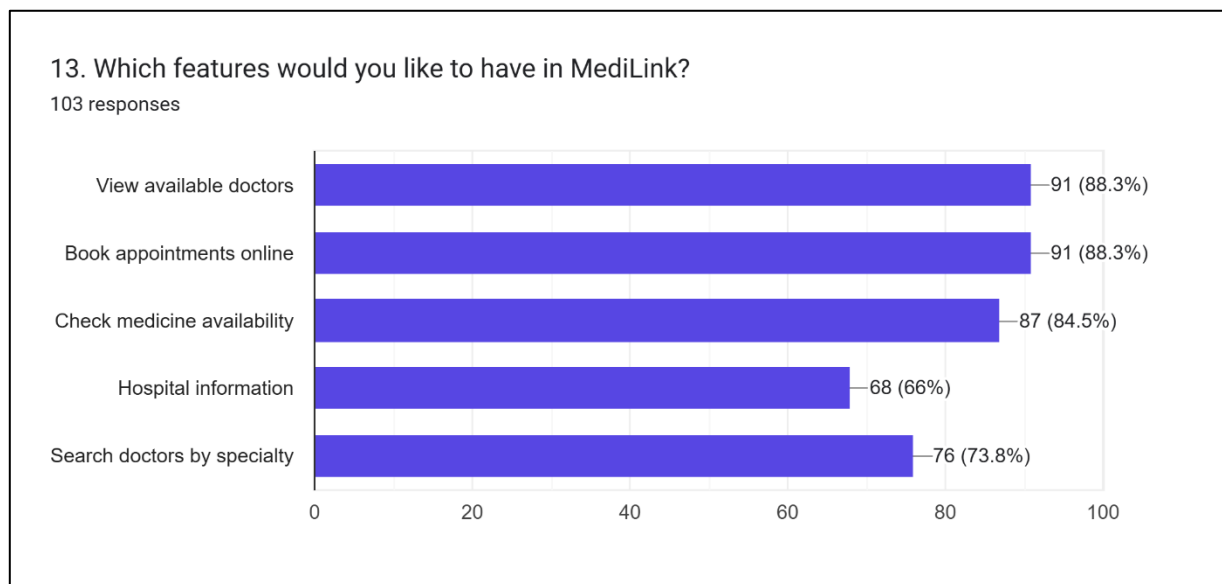
103 responses



Most Requested Features and Priorities

Proposed Feature	Respondents in Favour
View available doctors	88.3%
Book appointments online	88.3%
Check medicine availability	84.5%
Search doctors by specialty	73.8%
General hospital information	66%

Table 3: Feature preference ranking (n = 103)



Notably, the three highest-ranked features overall – viewing available doctors, booking appointments online and checking medicine availability – map directly onto the three capabilities retained in the revised MediLink scope and all three now exceed 84% support. This provides strong, data-driven justification for the scope reduction: the team is not simply building a smaller system but is prioritizing the exact features respondents value most.

When asked what matters most on a healthcare website in general, “easy appointment booking” was again the single highest-rated attribute at 82.5%, followed by “mobile-friendliness” (68.9%) and “secure payment” (64.1%), with “simple design” (60.2%) and “easy navigation” (49.5%) also rated highly. These preferences directly inform the interaction design and interface priorities described in later sections.

User Requirements and Expectations

Based on the questionnaire findings above, the following functional and non-functional requirements and user expectations have been identified for the revised, patient-focused MediLink platform.

Functional Requirements

- The system shall allow a patient to view a list of doctors together with their specialty, hospital and current availability.
- The system shall allow a patient to search or filter doctors by specialty and by hospital.
- The system shall allow a patient to select an available time slot and book an appointment with a chosen doctor.
- The system shall allow a patient to view, reschedule or cancel a previously made appointment.
- The system shall allow a patient to search for a specific medicine and view its stock status at a given hospital before travelling.
- The system shall allow a patient to enter their personal and appointment details and complete secure online payment for an appointment or service.
- The system shall provide the patient with confirmation (on-screen and where possible, via notification) after a successful booking or payment.

Non-Functional Requirements and Expectations

- **Usability:** given that 94.2% of respondents are already comfortable using healthcare websites but strongly favors “easy appointment booking” (82.5%) and “simple design” (60.2%), the interface must minimize the number of steps and avoid unnecessary complexity.
- **Accessibility for elderly users:** text must be large and readable and workflows must be forgiving of mistakes, since a portion of the primary and secondary user base is over 60 years of age.
- **Mobile responsiveness:** as 100% of respondents own a smartphone and 68.9% rated mobile-friendliness as important, the interface must be fully responsive and mobile-first.
- **Security:** since 64.1% of respondents rated secure payment as important, the payment workflow must use recognized secure payment handling and clearly communicate that transactions are protected.
- **Performance:** fast loading was rated important by a substantial share of respondents (43.7%); pages, in particular doctor and medicine search results, must return quickly.

- **Reliability of information:** medicine availability data must be kept reasonably current, since the core value of this feature is avoiding a wasted hospital visit.

Core Task Identification

Reviewing the questionnaire findings and the retained scope together, the team identified the core task that the MediLink patient platform must support: a patient locating an available doctor, securing an appointment with that doctor and completing payment for that appointment online optionally first checking medicine availability if the visit is medicine related. This core task subsumes and connects the three individual features defined in the revised scope, rather than treating them as unrelated, isolated functions.

Main Goal of the User

The main goal of the patient using MediLink is:

- “To secure a confirmed appointment with an available doctor and complete payment for it entirely online, with the option to first verify medicine availability, without needing to travel to the hospital or use the telephone.”

This goal is deliberately framed around the complete end to end outcome the patient cares about a confirmed, paid appointment rather than around any single screen or feature in isolation.

Breakdown of the Main Goal into Sub-Tasks

The main goal decomposes into four major sub-tasks, reflecting the natural sequence a patient follows when using the platform:

1. Register or log in to the MediLink platform.
2. Check the availability of doctors and select a suitable appointment slot.
3. Check the availability of a required medicine at the hospital
4. Enter patient and appointment details and complete an online payment to confirm the appointment

Actions Required to Complete Each Sub-Task

Sub Task 1: Register or Log In

- Open the MediLink website.
- Select “Login” if an account already exists or “Register” for a new account.
- Enter the required identity details.
- Verify the account.
- Land on the dashboard screen.

Sub Task 2: Check Doctor Availability and Book an Appointment

- Select “Find a Doctor” or “Book Appointment” from the dashboard.
- View the list of matching doctors along with their available dates and time slots.
- Select a preferred doctor and an available time slot.
- Review the appointment summary.
- Confirm the selection to move the appointment into a “pending payment” state.

Sub Task 3: Check Medicine Availability

- Select “Check Medicine Availability” from the dashboard or from within the appointment flow.
- Search for the required medicine by name.
- View the returned stock status (available /unavailable).
- Decide, based on the result, whether to proceed with, adjust or cancel the planned hospital visit.

Sub Task 4: Enter Details and Complete Online Payment

- Review and if necessary, update patient details.
- Review the appointment and fee summary.
- Select a payment method.
- Enter the required payment details.
- Submit the payment and wait for processing/authorisation.
- Receive an on-screen confirmation and, where available, a confirmation notification or receipt.

Hierarchical Task Analysis

The hierarchical task decomposition below follows standard HTA notation, in which task 0 represents the overall goal, numbered tasks (1, 2, 3...) represent sub-tasks and decimal-numbered items (1.1, 1.2...) represent the individual actions that make up each sub-task. Each numbered group is accompanied by a short “Plan” describing the order and conditions under which its child actions are carried out.

Task Hierarchy

5. 0. Book and pay for a hospital appointment online via MediLink
6. 1. Register or log in to MediLink
 1. 1.1 Open the MediLink platform
 2. 1.2 Enter login credentials or complete registration
 3. 1.3 Verify account
7. 2. Check doctor availability and select an appointment
 1. 2.1 Choose specialty
 2. 2.2 View list of available doctors and time slots
 3. 2.3 Select doctor and preferred time slot
 4. 2.4 Confirm appointment selection
8. 3. Check medicine availability
 1. 3.1 Search for medicine by name
 2. 3.2 Select hospital/pharmacy to check
 3. 3.3 Review stock status result
9. 4. Enter details and complete online payment
 1. 4.1 Review/update patient details
 2. 4.2 Review appointment and fee summary
 3. 4.3 Select payment method and enter payment details
 4. 4.4 Submit payment and receive confirmation

Plans

Plan	Description
Plan 0	Do 1, then 2. If the visit involves obtaining medicine, do 3 before proceeding. Then do 4 to complete the goal.
Plan 1	Do 1.1. If the patient already has an account, do 1.2 (login) and skip to 1.3 only if verification is requested; otherwise do 1.2 (register) then 1.3.
Plan 2	Do 2.1, then 2.2. Repeat 2.1-2.2 if no suitable doctor/slot is found. Do 2.3, then 2.4.
Plan 3	Do 3.1, then 3.2, then 3.3. If the result is “unavailable”, the patient may return to Plan 2 to reconsider the appointment before continuing.
Plan 4	Do 4.1, then 4.2. Do 4.3, then 4.4. If payment fails, return to 4.3.

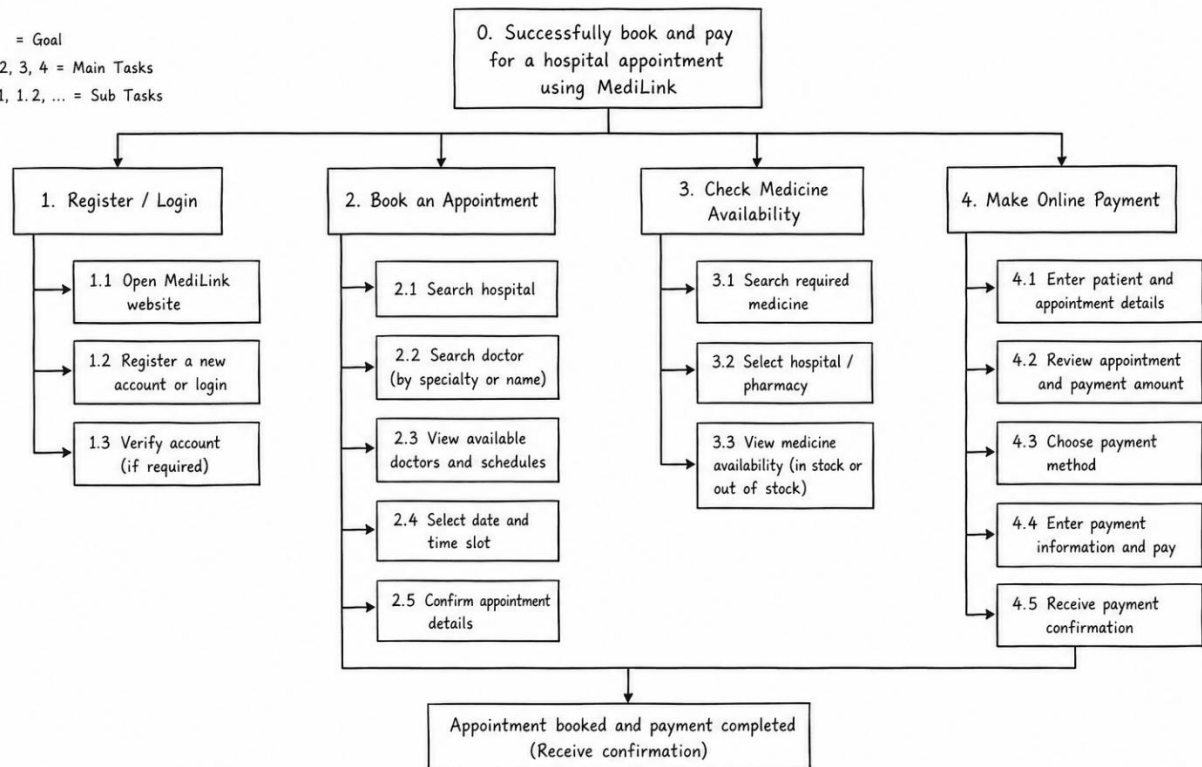
Table 4: Plans governing execution order within the hierarchical task analysis

Hierarchical Task Analysis (HTA) for MediLink

0 = Goal

1, 2, 3, 4 = Main Tasks

1.1, 1.2, ... = Sub Tasks



Conclusion

This report has re-examined the requirements collected for MediLink and used them to justify and support a deliberately reduced project scope a patient only web platform offering doctor availability and appointment booking, medicine availability checking and online payment. The questionnaire findings show strong, consistent support for exactly these three capabilities. They were the most requested features, address the most commonly reported problems (long waiting times, unavailable medicine, difficulty booking appointments) and align with respondents' expressed priorities of easy booking, secure payment and mobile friendly design.

By narrowing the scope, the team is able to focus design and development effort on delivering these three features to a high standard of usability, guided directly by the target users' documented needs and by the hierarchical task analysis presented above, which will inform the interaction design and interface prototyping in the next stage of the project.